THE OPEN SOURCE AUDIT

A Capstone Project for OSS NGMC course

Course: Open Source Software | Unit Coverage: 1 – 5 Max Marks: 100

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Chosen software: VLC media player

PROJECT REPORT:

Part A — Origin and Philosophy :

Origin of VLC Media Player

VLC Media Player was made in 1996 by students at École Centrale Paris. They called it VideoLAN Client. It was a project to stream videos over the school network.

- It had two parts: a server to send media and a client to play it.

In 2001 VLC Media Player became source. This meant developers, around the world could help make itbetter.

VLC Media Player got really good over time. It can play all kinds of audio and video. You do not need programs to make it work. VLC Media Player supports formats.

Philosophy of VLC Media Player

The development of VLC is guided by a strong set of principles:

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1. Open Source and Freedom

The VLC media player is free and open source. This means that the source code of the VLC media player is available for everyone to see. This approach encourages a things:

- Transparency
- Community contributions to the VLC media player
- Continuous improvement of the VLC media player

This is what the people behind the VLC media player believe in. They think that software should be free.

2. The VLC media player can Play Everything

- The people who made the VLC media player think that it should be easy to use. They believe in one idea:
- The VLC media player just works.
- The VLC media player can play any file you have like MP4 or MKV or AVI or MP3

.You do not need to install anything to make the VLC media player work. The VLC media player can even play files that are damaged or not complete. This makes the VLC media player very easy to use and very reliable.

3. The VLC media player Works on Many Platforms

The VLC media player is made to work on different platforms, such as

- Windows
- macOS
- Linux
- Android and iOS

This means that anyone can use the VLC media player no matter what device they have.

4. The VLC media player Has No Ads and No Spyware

The VLC media player is different from other free media players. The VLC media player does not have any ads. The VLC media player does not track what you do. The VLC media player does not have any hidden malware. The people behind the VLC media player care about your privacy.

5. The VLC media player is Made by a Community

The VLC media player is maintained by volunteers and developers from around the world. They work together through the organization. The decisions they make are not based on making money. They focus on making the VLC media player easy to use and innovative. They update the VLC media player regularly. Make it better.

6. The VLC media player is Simple but Powerful

The VLC media player has an interface but it can do many advanced things, such, as

- Streaming
- Converting file formats
- Synchronizing subtitles
- Recording media

The VLC media player is an useful tool.

A1. The problem this software was created to solve

A.VLC Media Player was made to fix some problems with playing videos and music on computers in the late 1990s:

1. Playing Kinds of Files Was Hard

Back then,

- Media players could only play a few kinds of files
- People had to install codes to play some files
- Sometimes files would not play because something was missing

B.VLC Media Player made this easier by putting all the codes it needed inside the player so people could play any kind of file without having to install anything extra.

2. Watching Videos Online Was Not Working Well

The main reason VLC Media Player was made was to:

- Send video over a computer network at École Centrale Paris
- Make sure videos played smoothly when sent over the network

C.VLC Media Player made it possible to send videos over the network and play them back in time which was hard to do with other tools.

3. Some Players Only Worked on Certain Computers

Before VLC Media Player:

- Some media players only worked on operating systems

This caused problems for people who used kinds of computers

D.VLC Media Player was made to work on many kinds of computers including Windows, Linux, macOS and mobile devices.

4. Using Media Players Was Frustrating

- People had to deal with:
- Complicated setup processes
- Videos that would not play correctly
- Players that would crash when trying to play some files

E.VLC Media Player made things simpler by making it easy to just open and play files.

5. Some Players Had Ads or Cost Money

- Some media players:
- Had ads or spyware
- Required people to pay for a license

F.VLC Media Player was free did not have ads. Was open-source, thanks, to the VideoLAN Organization.

VLC Media Player was made to solve the problems of playing different kinds of files sending videos over the network using different kinds of computers and making it easy for people to use so playing videos and music is simple and easy for everyone to do with VLC Media Player.

Here's a clear and simple A2 answer based on the license used by VLC:

A2. The License. What It Says

VLC Media Player is released under the GNU General Public License, a free software license that helps users and makes sure people can work together.

1. Four Freedoms of Free Software

The Free Software Foundation says there are four freedoms:

- You can run the program for any reason
- You can see how the program works and change it
- You can modify the program

- You can share copies with others

Does VLC's license give you all four freedoms?

- Yes it does. The GPL license used by VLC gives you all four freedoms.

2. Can a Company. Sell It Without Sharing Changes?

No they cannot keep changes secret.

- The GPL license makes sure changes are shared
- If someone changes VLC and shares it they must also share their changes

This helps keep improvements to everyone.

3. Difference Between GPL and MIT License

Feature	GPL License	MIT License
Type	Copyleft	Permissive
Source Code Sharing	Mandatory	Not required
Commercial Use	Allowed (with conditions)	Fully allowed
Restrictions	More strict	Very minimal

Key Difference:

- GPL makes you share changes
- MIT lets you keep changes private

4. "Free as in Free Beer" vs "Free as in Freedom"

- as in Free Beer → It is free but there may be rules
- Free as, in Freedom → You have rights to use, change and share

Example from VLC:

- VLC is free to download

- More importantly it is open-source and you can change it

The GPL license used by VLC Media Player helps users have freedom be open and work together making it a good example of free software.

A3.The Ethics of Open Source

Open source is more than a way of making software. It is about believing that knowledge should be shared with everyone improved by working together and made available to all people. A good example of this is the VLC Media Player, which shows how working together can help millions of people over the world.

1. Should All Software Be Open Source?

Argument For Open Source

- Open source makes things transparent so anyone can look at the code
- It helps people come up with ideas by working together
- It makes things more secure because problems can be found and fixed faster
- It makes technology available to everyone students and developers

For example VLC works on almost every platform because developers from all over the world contribute to it.

Argument Against Open Source

- Companies might not have a reason to come up with ideas if they cannot make money from them
- There is a risk that people might use or take advantage of the code in a bad way
- Some software, like security systems or special computer programs might need to be controlled
- Developers might not always get paid fairly for their work

Not all industries can work with an open model.

Balanced View

Not all software has to be source but the basic things that everyone uses and the tools that help people should be open.

2. Is It Ethical for Companies to Profit from Open Source?

Companies like Red Hat, Google and Meta make products using open-source software like Linux.

Ethical If:

- They give back to the community
- They make the ecosystem better by fixing problems giving money and making tools
- They follow the rules of the license

For example Red Hat contributes a lot to Linux and makes money by helping people use it.

Unethical If:

- They take advantage of the communitys work without giving back
- They take code change it and hide the improvements, which is not fair to the community

It is ethical only when companies give back to the community. Making money should not come at the cost of being fair to the community.

3. "Standing on the Shoulders of Giants" in Software

This phrase means building ideas on top of what we already know.

In software:

- Developers reuse libraries, frameworks and tools
- No one builds everything from scratch

Does Open Source Help or Hurt Innovation?

Enables Innovation

- Saves time by reusing existing code
- Lets developers focus on ideas instead of the basics
- Encourages trying things

For example VLC was built using existing ideas, about networking and multimedia.

Downsides

Relying much on open source might reduce deep understanding

Some developers might copy of create

Final Take

Open source greatly helps innovation. It does not replace coming up with ideas. It makes it happen faster.

- **Overall**

Open source is a way of thinking that is rooted in sharing working together and growing together. While it has challenges it is one of the powerful forces driving modern technology forward.

Part B — Linux Footprint

Installation method *— is it available via RPM, apt, or must it be compiled from source?*

- *On most LINUX based systems,VLC is directly available through package managers*

```
sudo apt update
sudo apt install vlc
sudo dnf install vlc
git clone https://github.com/videolan/vlc.git
cd vlc
./bootstrap
./configure
make
sudo make install
```

2. Key Directories

Type	Location	Purpose
Binary	`/usr/bin/vlc`	**Main executable**
Libraries	`/usr/lib/vlc/`	**Plugins & codecs**
Config (user)	`~/.config/vlc/`	**User settings**
System config	`/etc/vlc/`	**System-wide config (rare)**
Logs	`~/.local/share/vlc/logs/`	**User logs**

3. User and Group

- VLC always runs as the current logged in user and not a system service user
- No dedicated vlc user or group is created while using

WHY THIS MATTERS FOR SECUIRITY?
It limits damage if VLC crashes
It cannot access system files without the permission of the user.
It follows the principle of least privilege

4. Service Management

VLC is not a background service by default because it is a GUI/media application.

5. Update Model

VLC can be updated through linx package repositories

```
sudo apt update
sudo apt upgrade vlc
```

Part C — The FOSS Ecosystem

1. What VLC Media Player relies on

VLC Media Player needs a lot of tools and libraries to work properly. These are some of the ones:

- Core Multimedia Libraries
-
- FFmpeg is used for decoding and encoding audio and video
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- libavcodec handles the codecs
-
- libVLC is the main engine that VLC Media Player uses
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- System and User Interface Libraries
-
- Qt is used for the interface
-
- SDL is used for rendering multimedia
-
- Streaming
-
- Live555 is used for RTSP streaming
-
- OpenSSL is used for secure streaming

This means that VLC Media Player does not do everything on its own. It uses other powerful open-source components.

2. What VLC Media Player has made possible/Inspired

- VLC Media Player has influenced and helped many other tools.
- Libraries and Software Development Kits
- libVLC is used by developers to add video playback to apps
- Many apps on Android and iOS use VLC Media Players engine internally
- Custom versions of VLC Media Player
- There are custom versions of VLC Media Player for Smart TVs
- There are custom versions of VLC Media Player for set-top boxes
- There are custom versions of VLC Media Player for embedded systems
- Streaming and Media Tools
- VLC Media Player has inspired the creation of players and streaming tools
- VLC Media Player is used in education, broadcasting and surveillance systems

This shows that VLC Media Player is often used as a base technology not as a player.

3. How VLC Media Player relates to the LAMP Stack and the web

The LAMP Stack is made up of Linux, Apache, MySQL and PHP.
VLC Media Player is not a part of the LAMP Stack but it is connected to the web ecosystem.
Here is how VLC Media Player relates:

- VLC Media Player runs on Linux, which's part of the LAMP environment
- VLC Media Player can stream media over the internet acting like a small server
- VLC Media Player is used with web apps for things like video streaming and media processing
- For example a web app made with PHP can store videos and VLC Media Player can stream them over the network

So VLC Media Player acts as a layer for handling multimedia working alongside web stacks not inside them.

4. Community/Governance
There are platforms that the VLC Media Player community uses:

- Official forums, like the VideoLAN forums
- Mailing lists, where developers discuss things
- IRC channels for real-time chat
- Git repositories, where people can contribute to the code
- The VideoLAN Organization is in charge of VLC Media Player.

Some key things about the organization are:

- It is a -profit organization
- It is run by volunteers who develop the software
- Anyone from around the world can contribute to the code
- Decisions are made based on what's best for the technology not for making money

This is important because it means that VLC Media Player will be around for a long time and it will not be controlled by any one company. It also encourages people from, over the world to work together.
In conclusion VLC Media Player is part of an ecosystem of open-source libraries, tools and communities. It relies on projects and it also helps new ones get started making it an important part of the global open-source software network.

Part D — Open Source vs Proprietary

Dimension	Open Source (VLC Media Player)	Proprietary Alternative (Windows Media Player)
Cost	Completely free (no ads, no paid version)	Free with Windows, but tied to OS license
Security (Code Audit)	Source code is public → anyone can audit, find bugs, improve security	Source code is closed → only the company can audit
Support & Reliability	Community support, forums, documentation; very stable but no official customer support	Official support from Microsoft; reliable but limited customization
Freedom to Modify	Full freedom to modify and redistribute (GPL license)	No modification allowed; strict usage terms

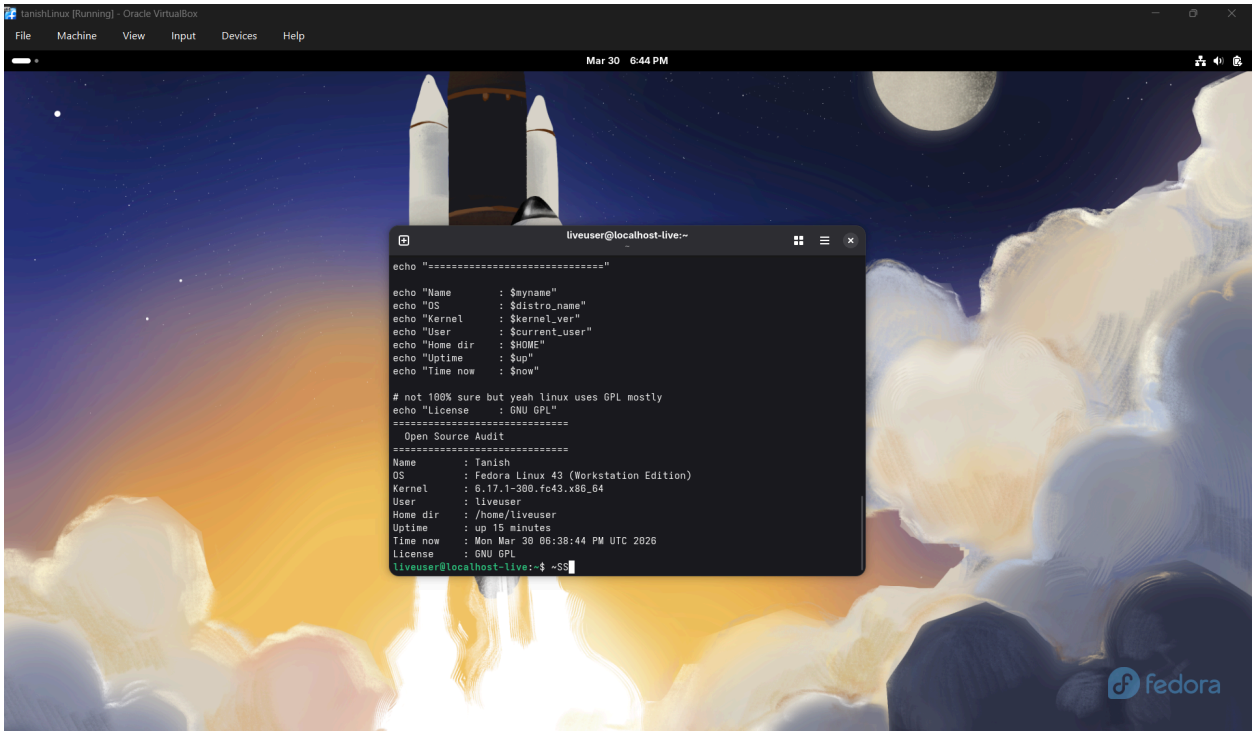
Community vs Corporate Control	Controlled by VideoLAN Organization (non-profit, community-driven)	Controlled entirely by Microsoft
Features & Compatibility	Plays almost all formats without extra codecs	Limited format support; may require codecs
Privacy	No tracking, no ads	May include telemetry depending on OS settings
User Experience	Simple but slightly less polished UI	More polished UI, better OS integration
Overall Verdict	Best for flexibility, compatibility, and freedom	Best for simplicity and tight Windows integration

Conclusion

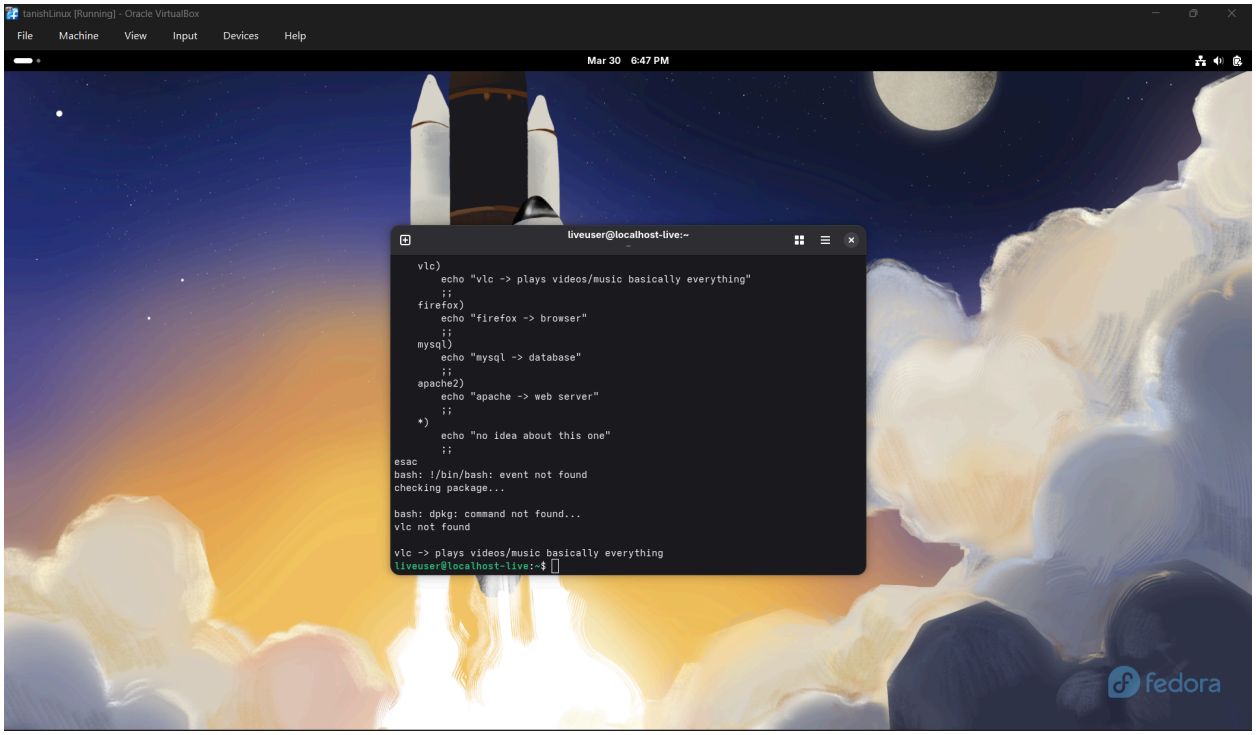
VLC Media Player is really good at playing media files. It works well in a lot of situations especially when you need to play types of files and you do not want to spend a lot of money. It is also good for people who care about keeping their information private. VLC Media Player can play any media file without needing extra help, which makes it very useful for students, people who make software and even professionals who test or stream media.
Some people might still like to use Windows Media Player because it looks nice and works with their computer system.. I think VLC Media Player is a good choice for most people. I like VLC Media Player because it is open and free and a lot of people work together to make it better. I would even like to help make VLC Media Player better because it's a project that anyone can contribute to.Even small things, like telling them about problems or making the instructions better can help. This helps keep the project going. It is a good way to work together to make technology better. I think this is a way to make progress and help people at the same time. VLC Media Player is a choice because it is flexible and reliable and it can be used in many different situations.

4. Shell Script Tasks

Script 1 — System Identity Report



Script 2 — FOSS Package Inspector



Script 3 — Disk and Permission Auditor
Script 4 — Log File Analyzer

Script 5 — Open Source Manifesto Generator